

KGML gasmet data processing

CO₂ N₂O and CH₄

Jesse Bealsburg & Katherine Bohn

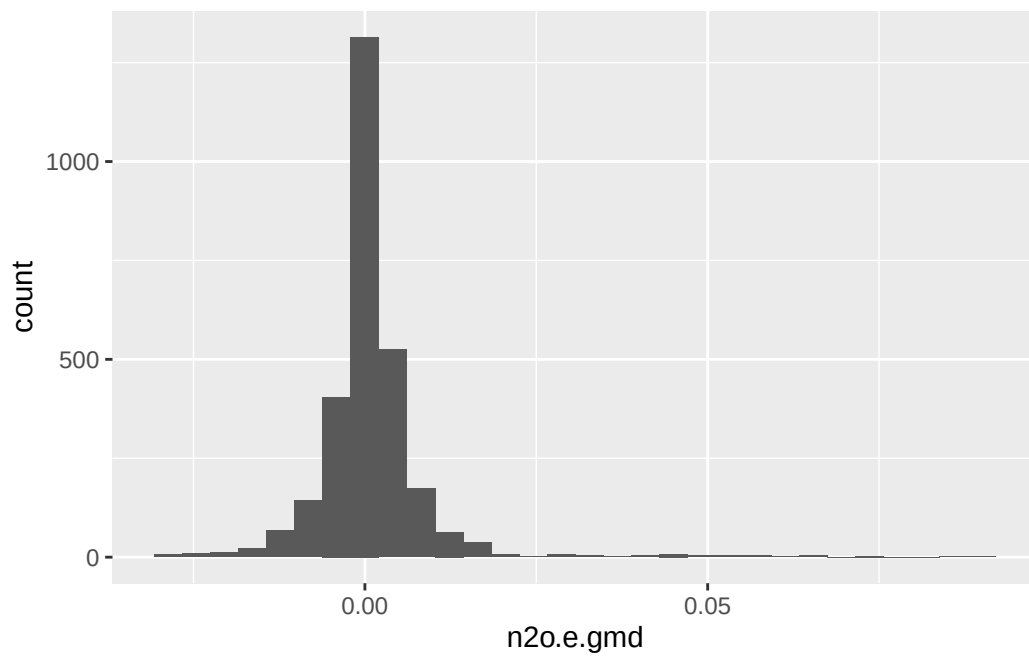
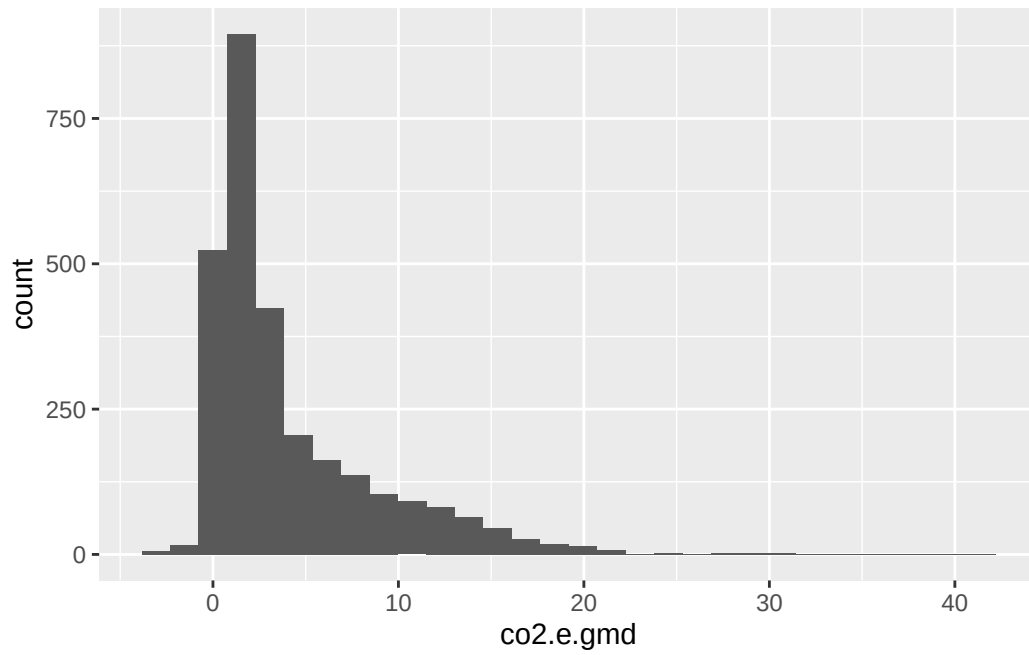
2025-04-04

Key points

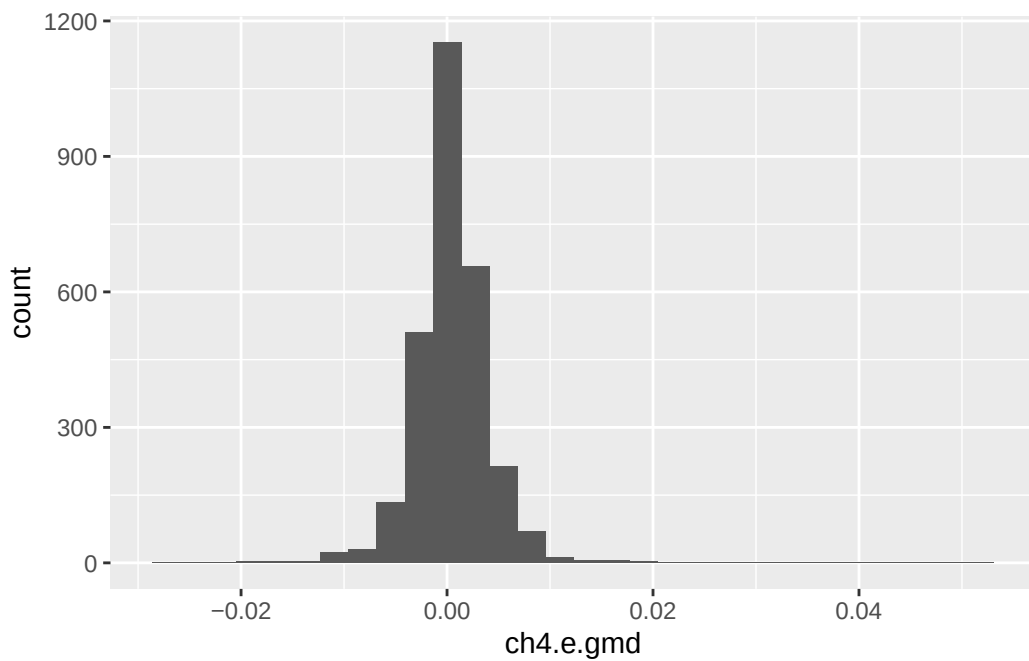
- *CO₂ N₂O and CH₄* converted from ppm to {elemental weight} × {unit of area} × {duration} like $\text{g} \cdot \text{m}^2 \cdot \text{day}^{-1}$
- Most *CO₂* fluxes are positive = good.
 - 105 of the 2836 predict positive *CO₂* emissions
- *N₂O* and *CH₄* fluxes are small and normally distributed around zero = concerning
- Kathryn independtly processed gasmet data. This data was brought in for comparison and the distribution of *N₂O* and *CO₂* remained = further investigation and discussion required
 - Is there a calibration issue for *N₂O* and *CH₄*?
 - Do we expect most N2O and CH4 data to be useless?

Visualization

Jesse



```
# A tibble: 149 x 5
# Groups:   experiment [5]
  experiment date      totalSlopeTally negSlopeTally negFluxPercent
  <fct>      <chr>          <int>          <int>          <dbl>
1 atat      2023-06-01           10             8            80
2 ncas      2024-10-08          28            20           71.4
3 ncas      2024-09-24          27            19           70.4
4 eddyflux  2023-05-24           10             7            70
5 ncas      2023-06-05          36            25           69.4
6 ncas      2024-04-09          16            11           68.8
7 eddyflux  2022-06-28          21            14           66.7
8 eddyflux  2024-07-29          12             8           66.7
9 ncas      2024-03-19          14             9           64.3
10 ncas     2024-04-25          14             9           64.3
# i 139 more rows
```



```
# A tibble: 149 x 5
# Groups:   experiment [5]
  experiment date      totalSlopeTally negSlopeTally negFluxPercent
  <fct>      <chr>          <int>          <int>          <dbl>
1 eddyflux  2023-07-05          12            11           91.7
2 eddyflux  2023-04-13          11            10           90.9
```

3	ncas	2023-04-13	11	10	90.9
4	atat	2022-09-27	20	15	75
5	oreimanure	2020-11-19	24	17	70.8
6	atat	2023-06-15	10	7	70
7	ncas	2023-07-05	16	11	68.8
8	ncas	2024-08-20	27	18	66.7
9	eddyflux	2024-10-28	24	16	66.7
10	ncas	2023-09-26	22	14	63.6

i 139 more rows

Kathryn's

I'd expect her process to be more robust than mine. Just ATAT and Eddyflux

